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Study on the responsiveness of U.S. stock market (S&P 500) to monetary expansion (M2) and real economic growth (Real GDP) across different monetary/fiscal policy eras.

Vilpert Nygård

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Author Vilpert Nygård

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Thesis supervisor Senior University Lecturer: Dr. Joni Kokkonen

Thesis advisor(s) Assistant professor: Christoph Huber

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Abstract

This thesis examines how U.S. stock market returns (S&P 500) relate to monetary expansion (M2) and real economic growth (real GDP growth) over 1960-2025 and assesses which variable provides greater explanatory content for stock return dynamics.

Using reduced-form vector autoregression (VAR) framework, the results indicate that neither M2 nor real GDP meaningfully predicts stock returns at quarterly horizons. At annual horizons, predictability becomes detectable, and real GDP exhibits stronger incremental explanatory content than M2 when both variables are included jointly, with the sign of the estimated relationship being negative.

Multiple-break testing using the Bai-Perron procedure finds no statistically significant structural breaks in the conditional-mean relationships, although subsample evidence suggests that any predictability weakens in the later part of the sample.

Keywords Stock return predictability, macrofinance, market dynamics, S&P 500, M2, Real GDP, vector autoregression, VAR, structural breaks, Bai-Perron

Tekijä Vilpert Nygård

Työn nimi Tutkimus Yhdysvaltain osakemarkkinoiden (S&P 500) reagointiherkkyydestä rahamäärän (M2) sekä reaalitalouden kasvulle (reaalinen BKT) eri raha- ja finanssipoliittisilla aikakausilla..

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Tiivistelmä

Tämä tutkielma tarkastelee Yhdysvaltojen osakemarkkinoiden tuottoja (S&P 500) suhteessa rahamäärän (M2) sekä reaalitalouden kasvuun (reaalinen BKT) 1960–2025 kattavalla tarkkailujaksolla, arvioiden kumpi tekijä selittää vahvemmin osakemarkkinoiden dynamiikkaa.

Käyttäen redusoitua vektoriautoregressiivistä mallia tulokset eivät lyhyellä, kvartaalisella aikavälillä osoita kummankaan muuttujan tilastollisesti merkittävällä tavalla selittävän osakemarkkinoiden tuottoja. Pidemmällä, vuosittaisella aikavälillä reaalinen BKT osoittaa suurempaa selittävää voimaa M2:een nähden, kun molemmat muuttujat sisällytetään samaan malliin. Arvioidun vaikutuksen suunta on negatiivinen.

Bai–Perron-menettelyyn perustuva useiden murtokohtien analysointi ei tuota tilastollisesti merkittäviä murtokohtia mallissa, vaikka osajoukkoanalyyseistä saadaan viitteitä havaitun vuorovaikutuksen heikkenemisestä otoksen loppupuoliskolla.

Avainsanat Osaketuottojen ennustettavuus, makrorahoitus, markkinadynamiikka, S&P 500, M2, reaalinen BKT, vektoriautoregressio, VAR, rakenteelliset murtumat, Bai-Perron

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1. Introduction

Financial markets are influenced by both liquidity conditions and real economic fundamentals. Prior research shows that monetary policy shocks affect equity prices (e.g., Thorbecke, 1995; Rigobon & Sack, 2003; Neuhierl & Weber, 2016), as does the expected growth of the real economy (Fama, 1990; Fichtner & Joebges, 2024). However, the relative importance of money growth versus real economic growth in explaining stock market movements remains a subject of debate. Moreover, major structural regime changes, such as the collapse of the Bretton Woods system, neoliberal policy reforms, and the evolving interaction between fiscal and monetary policy following the Global Financial Crisis, may have altered the responsiveness of financial markets over time.

This thesis therefore takes a comparative, long-term perspective on the effects of liquidity conditions and real economic growth on the U.S. stock market. By examining these relationships across different policy environments, it aims to contribute additional empirical evidence and to shed light for financial decision makers on which variable assumes a more dominant explanatory role under varying macroeconomic regimes.

1.1. Introduction to subject

The current global financial and economic environment is characterized by rapid and substantial policy shifts in liquidity provision, economic support, and prevailing economic doctrines. Examples include the large-scale liquidity programs following the 2008 financial crisis, fiscal support packages during the Covid-19 pandemic, and policy changes related to the reorganization of global trade under the Trump administration. At the same time, the world is experiencing significant technological transformation, such as advances in artificial intelligence and digitalization, that may substantially enhance economic output.

Against this backdrop, a detailed examination of the comparative effects of real economic growth and liquidity conditions on stock market returns is warranted in order to better understand the mechanisms through which equity returns are generated. The central research question of this thesis is therefore: Do liquidity conditions or real economic growth exert greater influence on future stock market returns?

This question lacks a clear scientific consensus, which is understandable given the inherent difficulty of comparing these two variables. Such an analysis requires numerous modeling choices and assumptions that can materially affect empirical outcomes. This thesis seeks to help fill this gap in the literature by adopting a simple and highly transparent empirical approach based on a reduced-form vector autoregression (VAR) framework. While this approach does not provide structural identification and may therefore limit causal interpretation, it offers clear and interpretable insights into the dynamic relationships among the variables of interest.

The thesis proceeds as follows. First, the relevant literature is introduced, synthesized, and expectations are derived based on prior findings (1.2. Literature review). This is followed by a brief overview of the main results and their alignment with the proposed expectations, intended to provide the reader with an initial yet thorough understanding of the study's outcomes (1.3. Summary of results). Readers seeking greater detail are encouraged to continue with the formal presentation of the research question and hypotheses (2. Research question and hypothesis), the data and methodology (3. Data and methodology), and the detailed results and conclusions (chapters 4. and 5.). For transparency, replicability, and for those interested in extreme detail, Appendix E (I) provides a link to a GitHub repository containing the code used for the methods and extended results.

1.2. Literature review

The interrelationships between real economic activity, inflation, as measured by changes in the Consumer Price Index (CPI), and stock market performance have long puzzled researchers. This section provides an overview of how existing literature conceptualizes and empirically assesses these relationships.

I (Real activity)

Let us start with real economic activity, which relationship with the stock market is rooted deep into economic theory. In one of the most fundamental stock valuation frameworks, the dividend discount model, denoted as:

$$P_0 = \frac{D_1}{r - g}$$

$$P_0 = value\ of\ asset\ at\ time\ 0, \qquad D_1 = first\ period\ dividend, \qquad r = discount\ rate,$$
$$g = dividend\ growth\ rate$$

We can see that the growth rate of the dividend is directly related to the subsequent stock price, as if we increase the growth rate of dividends g we end up with higher price P_0, and vice versa. The growth rate of dividends can be then directly linked to real economic growth, as the dividends of a firm are influenced by the cash flow it generates, which in turn is influenced directly by the level of economic activity. As generalized in a quote by Fichtner et al. (2024): "Theoretically, stock prices should move in line with economic activity, as the stock price of a firm should equal expected and discounted future payouts (mainly dividend payments) that should be linked to the firm's real activity."[1], while Campbell & Diebold (2005) further shows how stock returns embed expectations of future growth.

Despite this strong theoretical foundation, empirical evidence suggests that the relationship between real economic growth and stock market returns is relatively weak in the short run. Fichtner et al. (2024) show that real economic growth exhibits only minor contemporaneous correlation with stock market returns, with the relationship strengthening primarily over longer time horizons. This observation is consistent with earlier findings by Fama (1990), who shows that approximately 43% of the variance in stock returns can be attributed to news about expected cash flows, implying that a substantial portion of return variability originates from other sources.

A closer examination of the dividend discount framework highlights the critical role of the discount rate r. Campbell and Shiller (1988) formally decompose stock price movements into components related to expected future cash flows and expected future returns (discount rates). This decomposition underscores that fluctuations in discount rates can dominate short-term stock market dynamics, potentially obscuring the influence of real economic activity at higher frequencies.

Taken together, these findings suggest that while real economic activity should be positively related to stock market returns, this relationship is likely to be highly sensitive to the time horizon considered. In the short run, the signal from real activity may be overwhelmed by volatility in discount rates and other financial factors. Over longer horizons, however, the influence of real economic growth becomes more pronounced and may approach statistical significance, consistent with the results of Fichtner et al. (2024) and Fama (1990).

Recent evidence further suggests that this relationship may have weakened or structurally changed since the turn of the millennium. Fichtner et al. (2024) find indications of a break in the correlation between real economic activity and stock returns around this period, possibly reflecting financialization, repeated

[1] Fichtner et al. (2024) refers here to Shapiro (1988).

macroeconomic shocks, or evolving policy regimes. Accordingly, differing results may be expected when comparing subsamples before and after the millennium, and the presence of heteroskedasticity is also plausible.

Overall, despite these complications, there remains substantial theoretical and empirical support for a positive long-run relationship between real economic activity and stock market returns. In this thesis, this relationship is henceforth referred to as the real activity channel.

II (Inflation, liquidity, and monetary policy)

We begin by considering the relationship between inflation, commonly denoted by changes in the Consumer Price Index (CPI), and stock market performance. Empirical evidence across countries and time periods, such as Sharpe (2000) and Zhang (2021), generally finds a negative correlation between inflation and stock returns. In line with standard economic theory, rising inflation tends to coincide with lower stock returns, while disinflation or falling prices are often associated with higher returns.

At first glance, this appears to contradict a substantial body of literature showing that expansionary monetary policy actions, such as interest rate cuts or liquidity-providing measures, are positively correlated with stock market returns (e.g, Thorbecke, 1995; Rigobon & Sack, 2003; Neuhierl & Weber, 2016; Sharif, Bouteska, Abedin, Cotturone, 2025). Resolving this apparent contradiction requires a clear distinction between inflation outcomes and policy actions affecting liquidity conditions.

A useful starting point is the Quantity Theory of Money (QTM)[2], expressed as:

$$MV = PY$$

$$M = quantity\ of\ money, \qquad V = velocity\ of\ money, \qquad P = price\ level, \qquad Y = real\ GDP$$

While liquidity conditions influence inflation through changes in M and V, increases in real output Y can offset upward pressure on prices. As such, expansionary monetary/fiscal policy does not mechanically translate into higher inflation, particularly in the presence of slack or productivity gains.

Fama (1981) provides more formal reasoning through the proxy hypothesis, arguing that inflation does not causally affect stock returns. Instead, inflation serves as a proxy for expected real economic activity, where periods of high inflation are typically associated with expectations of weaker future real growth, which depresses stock prices. Under this view, inflation captures expectations about real activity rather than exerting an independent effect on equity returns. Therefore, with this interpretation, inflation itself is not treated as a primary causal driver of stock returns in this thesis. Instead, the focus shifts to liquidity conditions, largely influenced by monetary policy, which are more directly linked to asset pricing dynamics.

The literature on monetary/fiscal policy and stock returns largely converges on a positive relationship. Thorbecke (1995), for example, finds that expansionary monetary policy shocks lead to positive industry-level stock return responses, controlling for inflation-related effects following the framework of Boudoukh, Richardson, and Whitelaw (1994). Sharif et al. (2025) similarly documents that interest rate cuts and liquidity programs are associated with rising stock returns, although they also note periods where the relationship reverses, which are attributed primarily to long sample periods, structural change, and heteroskedasticity within used frameworks.

Neuhierl et al. (2016) further demonstrate that monetary policy affects stock returns not only through realized actions but also through communication and tone, which shape expectations about future policy

[2] The origins of this theory are not fundamentally known, having its roots around the renaissance period. For the purpose of this thesis the work of Friedman (1987) is used as reference.

paths. Despite methodological differences, these studies consistently support the conclusion that increases in expected liquidity conditions are associated with higher stock returns.

In summary, actions that improve liquidity conditions are expected to have a positive long-run relationship with stock market returns. This relationship is referred to throughout this thesis as the liquidity channel.

III (Comparing effects of real activity and liquidity conditions)

Research directly comparing the effects of real economic activity and liquidity conditions on stock market returns remains limited. This scarcity is not surprising, as these variables are highly interdependent, time-varying, and strongly influenced by expectations. Moreover, clean identification is challenging due to data limitations and feedback effects.

The interdependence among real activity, liquidity conditions, and stock market returns gives rise to multidirectional causality. Not only do real activity and liquidity influence stock returns, but stock returns themselves also affect real economic outcomes and policy decisions. Higher stock returns can stimulate capital expenditure and investment, thereby supporting future real activity, a relationship theorized and empirically supported by Fama (1981). Similarly, Rigobon et al. (2003) show that rising equity markets increase the likelihood of monetary policy tightening, while falling markets are associated with policy easing. This implies that stock returns may be negatively correlated with future liquidity conditions.

The interaction between real activity and monetary policy is among the most established relationships in macroeconomic theory. Keynesian frameworks[3] imply that economic downturns warrant looser liquidity conditions, while periods of strong growth call for tightening. This logic remains embedded in modern policy practice, such as the Federal Reserve's dual mandate of maximum employment and price stability (FED, 2021). The causal link from liquidity to real activity is also well supported: increased liquidity lowers financing constraints, boosts investment, and stimulates output, as shown theoretically by Keynes (1936) and empirically by Beck, Döttling, Lambert, van Dijk (2023).

These intertwined relationships imply severe collinearity challenges for empirical models attempting to estimate the effects of two, or all three variables simultaneously. While not insurmountable, these challenges complicate interpretation and demand careful methodological choices, which are discussed further in the VAR subsection below.

In addition, the relationships among these variables are time-varying. Stock return volatility is well known to be unstable over time (Schwert & Seguin, 1990; Whitelaw, 1994), and similar time variation has been documented for liquidity conditions and real activity (Justiniano & Primiceri, 2008; Avramov, Cheng, and Hameed, 2016). This further complicates identification and limits the usefulness of simple linear regression approaches.

Most existing studies therefore rely on vector autoregression (VAR) frameworks to model these dynamics. While VAR is well suited to capturing interdependencies, it can generate anomalies and sensitivity to specification choices, making firm conclusions difficult. Furthermore, results from studies focusing on only one variable, either real activity or liquidity, are often not directly comparable due to differences in controls, identification strategies and timeframes.

One notable attempt to compare these effects directly is Cassola &Morana (2002), who employ a structural vector error correction model (SVECM) and find that productivity shocks dominate stock market movements in the long run, while monetary policy appears to respond more to the market than vice versa. However, this study relies on a relatively short, and solely shorter term quarterly European sample (1980–2000), with only 80 datapoints and predates many major structural changes in global financial markets. Moreover, evidence

[3] Here referring to Keynes (1936)

suggests that the U.S. economy plays a uniquely dominant role in global dynamics and is less affected by external conditions than other economies (Fichtner et al. 2024), motivating a U.S. focused analysis for better identification of core effects.

IV (VAR)

Given the multidirectional and dynamic relationships described above, an econometric framework capable of modeling joint endogeneity is required. The VAR[4] model is therefore a natural choice, as it treats all variables in the system as jointly endogenous and captures their dynamic interactions through lag structures.

A VAR can be understood as a system of linear equations in which each variable is regressed on its own past values and the past values of all other variables in the system. This approach allows for flexible modeling of feedback effects without imposing strong prior structural assumptions. The technical details of the VAR implementation used in this thesis are discussed in the 3.2 Methodology section.

This thesis closely follows the methodological approach of Sharif et al. (2025), which provides a recent and transparent application of VAR methods to the study of monetary policy and stock returns, while openly acknowledging potential limitations and anomalies. Although the SVECM framework used by Cassola et al. (2002) could offer stronger identification, its limited use in other closely related literature and its reliance on strong identifying assumptions make it less suitable for the objectives of this study. The VAR employed in this study offers a balance between empirical flexibility and interpretability, allowing clearer inference while maintaining consistency with broader literature.

V (Conclusions)

In summary, the literature provides substantial evidence that both real economic activity and liquidity conditions influence stock market returns. These relationships are neither stable nor unidirectional, but they tend to emerge more clearly over longer time horizons. Importantly, stock returns themselves feed back into both real activity and liquidity conditions, giving rise to what this thesis refers to as anticipation channels[5] for real activity and liquidity.

Despite extensive research on each channel individually, relatively little work has been done to assess their comparative importance within a unified framework. Whether real economic activity or liquidity conditions constitute the primary driver of stock market returns therefore remains an open question. This gap in literature provides a clear motivation for the empirical analysis undertaken in this thesis.

1.3. Summary of results

This chapter provides a concise summary of the empirical findings of the thesis. Using reduced-form VAR methods, Granger-causality tests, impulse response functions (IRFs), and forecast error variance decompositions (FEVDs), the analysis evaluates whether U.S. stock market returns (S&P 500) are related to real economic activity (real GDP growth; RGDP) and monetary expansion (M2 growth), and whether these

[4] Main references here: Fama (1990), Fichtner et al. (2024), Neuhierl et al. (2016), Rigobon et al. (2003), Sharif et al. (2025), and Thorbecke (1995).

[5] These 'channel' terms are not much used in official research and are mainly common in informal discussion. I find them valuable to more simply refer to relationships between the discussed variables.

relationships operate differently across horizons and regimes. Results are reported for quarterly data (full and outlier-removed) and an annual dataset constructed to emphasize longer-horizon dynamics.

Structural breaks

Structural break testing using the Bai-Perron methodology yields no statistically significant evidence of breaks in the estimated relationships across the tested specifications, and across datasets (quarterly and annual). When the model is forced to include breaks, candidate dates recur around 1977–1978, parts of the early-2000s to GFC window, and occasionally 2008–2009, but these do not achieve statistical significance. The baseline interpretation is therefore that the relationships are sufficiently stable in conditional-mean terms to justify full-sample estimation, while acknowledging that gradual drift may still exist.

Macro-to-market relationships

Evidence for a predictive relationship from macro variables to stock returns is primarily horizon-dependent, with:

- **Quarterly frequency**: Neither RGDP nor M2 exhibits robust predictive power for S&P 500 returns. Coefficients in the S&P 500 return equation are generally small and statistically insignificant, and Granger causality tests typically fail to reject the null of no predictability. Removing Covid-era outliers marginally improves some estimates (occasionally approaching significance at weaker thresholds), but the results remain unstable and not robust at the 5% level.
- **Annual frequency**: In contrast, the annual dataset yields clearer evidence of predictability. Both RGDP and M2 show statistically significant predictive content for returns in separate (bivariate) settings, supported by Granger-causality testing. However, in both cases the estimated relationship is negative, contradicting the simplest theoretical expectation of a positive long-run effect. Subsample checks suggest that these relationships are stronger earlier in the sample and weaken in later periods.

Overall, the results indicate that any macro-to-market link is more detectable at longer horizons and is likely obscured at quarterly frequency by return volatility and expectation-driven dynamics.

Anticipation channels

The analysis also evaluates whether stock market returns predict subsequent macro variables:

- **Returns → RGDP**: The quarterly VAR provides consistent evidence that S&P 500 returns contain predictive information for RGDP growth. The lagged S&P 500 term in the RGDP equation is statistically significant (small and positive), and Granger causality tests reject the null of no predictability in the quarterly data. However, when moving to annual frequency, this predictability largely disappears, coefficients become insignificant and Granger causality is not supported, suggesting the effect is more shorter-term.
- **Returns → M2**: Evidence for returns predicting M2 is weak. In the full quarterly sample, S&P 500 lags in the M2 equation are generally insignificant and Granger-causality tests do not reject the null. Some episodic significance appears in narrower time windows and is sensitive to crisis periods. Removing Covid-era outliers reduces apparent predictability further, suggesting that any relationship is not stable and may be driven by extreme observations rather than persistent dynamics. Annual data similarly fails to show robust predictability from returns to M2.

In short, the anticipation channel is supported for real activity at short horizons (quarterly), but not for broad monetary expansion.

Comparative importance of RGDP versus M2

The core comparative analysis places RGDP and M2 jointly in a trivariate VAR and evaluates which variable carries greater explanatory importance for stock returns:

- **Quarterly joint model**: Neither variable exhibits meaningful predictive power for S&P 500 returns. Coefficients are insignificant, and Granger causality, IRF, and FEVD outputs do not support strong macro-to-market transmission at quarterly horizons. The more consistent significant dynamics in the quarterly system occur outside the return equation (e.g., links among RGDP and M2 and the returns → RGDP channel).
- **Annual joint model**: In the annual system, RGDP dominates M2. RGDP retains statistically significant predictive content for returns, while M2 loses significance once RGDP is included. IRFs show a clearer and more pronounced return response to RGDP shocks than to M2 shocks, and FEVD attributes a larger share of forecast error variance in returns to RGDP than to M2 at longer horizons. Subsample checks again indicate that this dominance weakens in later periods, consistent with broader evidence of instability.

The overall conclusion from the comparative analysis is that, within the framework and sample of this thesis, real economic growth provides more incremental information about long-horizon return dynamics than broad monetary expansion, while both are largely uninformative for short-term return prediction.

2. Research question and hypotheses

Based on the literature review, the comparative effects of real economic growth and liquidity conditions on stock market returns appear insufficiently studied. This thesis therefore examines whether indicators of real activity and liquidity conditions contain predictive information about future U.S. stock market returns and, where liquidity conditions are going to be proxied by monetary expansion[6], therefore more broadly, how realized return dynamics can be interpreted in differing macroeconomic and policy environments.

The empirical strategy proceeds in four steps. First, I test for evidence consistent with the real activity channel and the liquidity channel discussed in the literature review. This is done by estimating models in which stock market returns are individually related to (i) real economic activity and (ii) monetary expansion, and by evaluating whether these predictors exhibit statistically significant predictive content for returns, particularly over longer horizons. In addition to coefficient-based inference, I use Granger-causality[7] tests to assess whether lagged values of each macro variable improve forecasts of stock returns within the chosen system.

Second, I test for evidence consistent with anticipation channels, i.e., whether stock market returns contain predictive content for subsequent real activity and monetary expansion. This step is motivated by the literature emphasizing multidirectional linkages and policy feedback. The anticipation channels are assessed using the same system-based framework and complementary Granger-causality testing, with the goal of evaluating whether the data supports multidirectional dynamics and thereby justifying a VAR-based approach.

Third, I examine whether these relationships are stable over time by testing for structural breaks using the Bai-Perron methodology. The purpose is to identify statistically supported break dates in the predictive

[6] This is a practical choice derived from the QTM discussed in the literature review, which denotes liquidity conditions as quantity and velocity of money. Limiting implications of this are discussed in 5. Conclusions chapter.

[7] This test has been criticized for the use of the 'causality' term. More on this in the 3.2. Methodology subsection and Appendix A.

relationships, and to assess whether the estimated dynamics differ meaningfully across regimes that may correspond to changes in policy frameworks or macroeconomic conditions.

Finally, I conduct a comparative analysis by including both real activity and monetary expansion within the same multivariate system. This joint specification allows assessment of (i) statistical significance and effect sizes conditional on the other variable, (ii) whether Granger-causality evidence changes once both predictors are included, and (iii) how the relative contribution of each variable evolves over the forecast horizon using forecast error variance decomposition (FEVD) and Impulse Response Function (IRF).

2.1. Justification of research

This study is motivated by the limited amount of research directly comparing real economic activity and monetary expansion as joint predictors of stock market returns. The thesis contributes by providing up to date evidence on the individual relationships between these variables and the stock market, a topic that remains debated and potentially time-varying. For example, recent work suggests that the stock-growth relationship is unstable and may differ across periods (Fichtner et al. 2024)

While this thesis does not aim to produce a mechanical forecasting rule for equity markets, it seeks to provide structured evidence on how stock returns co-move with real activity and liquidity indicators across time and regimes. Such insight may be particularly relevant in an environment characterized by shifts in monetary/fiscal policy frameworks and rapid technological change.

From a finance perspective, the results may help financial decision-makers and investors better understand how macroeconomic growth and monetary expansion relate to equity return dynamics, improving interpretation of market behavior and the formation of expectations. It also could allow these actors to focus their time and effort into examining the side of the equation inferred more prominent by the results, thus allowing for more effective and better use of resources, when for example making investment decisions.

2.2. Hypotheses

To address the research question, the following hypotheses are tested:

$H_I = Real\ activity\ channel\ exists$

$H_{II} = Liquidity\ channel\ exists$

$H_{III} = Anticipation\ channel\ for\ real\ economy\ exists$

$H_{IV} = Anticipation\ channel\ for\ liquidity\ exists$

$H_V = Data\ has\ structural\ breaks\ for\ both\ variables$

$H_{VI} = GDP\ shows\ dominant\ explanatory\ power\ over\ M2$

Hypothesis I is based on the expectation that real economic growth is positively related to stock returns over longer horizons (e.g., Fama, 1990; Campbell et al., 2005; Fichtner et al., 2024). Real activity is proxied by Real GDP growth (RGDP). The hypothesis is evaluated by (i) examining the relevant coefficients in the return equation within the chosen VAR framework and their statistical significance, and (ii) conducting a

Granger-causality test with the null hypothesis that RGDP does not Granger-cause S&P 500 returns. All tests will be conducted at the 5% significance level.

Hypothesis II is tested similarly, with monetary expansion proxied by M2. The expectation of a positive relationship between liquidity-improving policy conditions and stock returns is grounded in the monetary policy and asset pricing literature (e.g., Thorbecke, 1995; Rigobon et al., 2003; Neuhierl et al., 2016) and is consistent with recent VAR evidence over long samples.

Hypothesis III examines whether stock market returns predict subsequent real activity. Using the same system framework, I test whether lagged S&P 500 returns exhibit predictive content for RGDP growth, and I perform the corresponding Granger-causality test with the null that S&P 500 returns do not Granger-cause RGDP growth. Hypothesis IV applies the same logic to M2, testing whether returns Granger-cause monetary expansion and whether return lags are significant predictors in the relevant equation.

Hypothesis V follows the literature emphasizing instability and regime dependence in return predictability and macro-financial relationships, motivating tests for multiple structural breaks (e.g., Smith & Timmermann, 2021). This is evaluated using the Bai-Perron method to detect and date statistically significant breaks in the estimated relationships. I do not impose specific break dates ex ante, since the literature does not pinpoint a single agreed set of breakpoints across samples and specifications.[8] Alternatively dates of prominent policy shifts could be used, but I determine choosing which policy shifts to include and when to account for them, to be outside the scope of this study.[9]

Hypothesis VI is motivated by the view that real-side shocks can dominate long-run stock market dynamics in multivariate macro-financial systems, as suggested in the euro-area literature emphasizing the role of productivity/real shocks for equity movements (Cassola et al. 2002). In the joint model including both RGDP and M2, dominance is assessed using (i) coefficient magnitudes and statistical significance, (ii) Granger-causality tests for each predictor conditional on the other, (iii) impulse response analysis (with bootstrapped confidence intervals), and (iv) FEVD to compare the share of forecast error variance in S&P 500 returns attributable to RGDP versus M2 across horizons.

3. Data and methodology

As previously discussed, this thesis evaluates dynamic relationships among three U.S.-based macro-financial concepts: real economic activity, monetary expansion/liquidity conditions, and stock market returns. Because these are theoretical constructs rather than directly observed numerical variables, empirical analysis requires the use of appropriate proxies. The proxies used in this thesis are as follows.

[8] For example, Fichtner et al. (2024) only remarked that the relationship between real economic growth and stock market returns breaks down somewhere after the 1980s (based on conclusions from previous studies), nor is the work of, for example Smith et al. (2021) relevant enough for my subject to conclude any days.

[9] Policy shifts are extremely hard to pinpoint to certain dates, as they cannot be just viewed to have happened the date of enactment. This because the markets react to these predictively. For example, a bill going through congress is well knows a long time before it passes to law, and the market prices in the effect of this bill relative to its passing probabilities already in this time. Furthermore, often policy shifts are associated strongly to certain individuals, whose behavior might already be priced in prior to election.

Stock market returns are proxied by returns on the S&P 500, which is widely used as a benchmark for large-cap U.S. equities and covers approximately 80% of available U.S. equity market capitalization[10], and it has been relied upon as the 'market portfolio' or 'benchmark index' in numerous studies, including the likes of Neuhriel et al. (2019) and Rigobon et al. (2003). Henceforth, this variable is referred to as S&P 500.

Real economic activity (real growth) is proxied by Real Gross Domestic Product (Real GDP) from the U.S. Bureau of Economic Analysis as disseminated through FRED (series GDPC1). FRED defines real GDP as "the inflation adjusted value of the goods and services produced by labor and property located in the United States." (FRED. 2025A). This variable is henceforth referred to as Real GDP or RGDP.

Monetary expansion / liquidity conditions are proxied by M2 from the Board of Governors of the Federal Reserve System (as disseminated through FRED, series M2SL), which consist of the following: "Before May 2020, M2 consists of M1 plus (1) savings deposits (including money market deposit accounts); (2) small-denomination time deposits (time deposits in amounts of less than $100,000) less individual retirement account (IRA) and Keogh balances at depository institutions; and (3) balances in retail money market funds (MMFs) less IRA and Keogh balances at MMFs." (FRED. 2025B)[11] This proxy is henceforth referred to as M2. As an additional note, M3 is not used because the Federal Reserve discontinued publication of M3 in March 2006, which prevents consistent measurement through the full sample.

3.1. Data

This section describes the data sources, frequency choices, transformations, and robustness-related adjustments.

Real GDP (RGDP). RGDP is obtained from the FRED database (Federal Reserve Bank of St. Louis), sourced from the U.S. Bureau of Economic Analysis (series GDPC1). The series is quarterly, measured in billions of chained 2017 dollars, and reported as a seasonally adjusted annual rate. To ensure stationarity and interpretability, RGDP is converted from levels into a growth-rate series, measured as percentage change from the previous observation. The estimation sample is set to 1960Q1–2025Q1[12], yielding 261 quarterly observations (noting that growth-rate construction requires the preceding quarter). Appendix B (Graph I) plots RGDP growth over time. (FRED, 2025A)

Money supply (M2). M2 is obtained from FRED (series M2SL), sourced from the Board of Governors of the Federal Reserve System. The original series is monthly, measured in billions of dollars, and seasonally adjusted. Because RGDP is quarterly, M2 is converted to quarterly frequency using an end-of-quarter aggregation (i.e., the last available month in each quarter). M2 is then expressed as quarterly percentage change from the previous quarter, consistent with RGDP. The estimation sample is 1960Q1–2025Q1 (261 observations). Appendix B (Graph II) plots the M2 series over time. (FRED, 2025B)

The data for the S&P 500 variable will be gathered from two different data sources. The first and primary one of these is the Center for Research in Security Prices (CRSP), from which daily returns (per closing price) data of the S&P 500 index returns is gathered for the timeframe of 3.7.1962 to 31.3.2025. I compound these daily returns into quarterly format using the formula:

[10] This is of course a constantly changing value and a rough approximation, I refer here to Dow Jones (2025).
[11] Post May 2020 the definition changes slightly, but this change doesn't numerically affect the M2 variable and mainly affecting the M1 variable providing a reason for choosing M2 instead of M1.
[12] Sample size is chosen as the largest possible allowed by the limiting factor of M2 data.

$$R_{quarter} = \prod_{t=1}^{n}(1 + r_t) - 1$$

and thus, get quarters Q3-1962 to Q1-2025. This proves a problem as to account for as long of a timeframe as possible, I would like to continue the data back to 1959. For this purpose, I have decided to use an additional data source to fill in this gap. Yahoo finance was chosen due to its easily available, clear, and back checkable quarterly S&P 500 returns data. To further make sure the two sources were comparable, I crosschecked the quarterly data from Yahoo finance to the quarters formulated from CRSP data and determined them to be matching. Appendix B (graph III) shows the plotting of this data. (CRSP, 2025; Yahoo, 2025)

Stationarity testing. Stationarity is important for linear VAR inference. Classic evidence shows that many macroeconomic level series behave as non-stationary processes, which can lead to spurious regression and invalid inference when modeled in levels (Nelson & Plosser, 1982). For this reason, each transformed series is tested using four common unit-root/stationarity tests: ADF, Phillips–Perron (PP), KPSS, and ADF–GLS[13]. Overall, the results strongly support stationarity of the transformed series; full test outputs are reported in Appendix C.

Preliminary association and horizon choice. Appendix B (Graphs IV–V) displays scatter plots of RGDP and M2 against S&P 500 returns. Visually, the contemporaneous relationship appears weak, and simple linear fits yield very low explanatory power. This is consistent with the literature review's emphasis that macro-financial relationships may be horizon-dependent and mediated by expectations, motivating lag-based modeling (VAR/Granger) rather than contemporaneous linear regression.

Yearly dataset. To examine longer-horizon dynamics more directly, an additional yearly dataset is constructed from the quarterly series. Annual returns/growth are formed by compounding within each calendar year by the compounding formula displayed earlier. The resulting yearly sample runs from 1961 to 2024, where the observation labeled 1961 corresponds to growth/return accumulated from 1960 to 1961, yielding 64 annual observations.

Outliers and robustness. Appendix B (Graphs IV–V) indicate three notable outliers, around the early-2020 Covid-19 shock. These observations reflect extreme but economically meaningful dynamics. The baseline analysis retains these observations to preserve the full distribution of macro-financial conditions. For robustness, an additional dataset excluding these outliers' is constructed and used for sensitivity checks. For the annual dataset, these outliers are substantially dampened by aggregation and are therefore less influential.

3.2. Methodology

This section describes the econometric methods used to test the six hypotheses introduced in 2.2. Hypothesis section. The aim is to provide an intuitive overview of the core models and tests while referring more technical derivations to standard references and appendices where appropriate.

I (VAR)

The central empirical framework of this thesis is the vector autoregression (VAR) model, which is widely used to study dynamic interactions among macro-financial variables. The VAR framework is particularly

[13] References for the test as follows: Dickey & Fuller (1979), Phillips & Perron (1988), Kwiatkowski, Phillips, Schmidt, Shin (1992), and Elliot, Rothenberg, Stock (1996).

suitable here because the literature review suggests multidirectional relationships among stock returns, real activity, and liquidity conditions/monetary expansion. Key methodological references include Sims (1980) and Lütkepohl (2005), alongside applied work in macro-finance using VAR-based identification and interpretation tools (Sharif et al. 2025; Thorbecke, 1995; Neuhierl et al. 2016; Fichtner et al. 2024).

To build intuition, consider a standard linear regression:

$$Y_t = \alpha + \beta X_t + \varepsilon_t,$$

which models a dependent variable Y_t as a function of an explanatory variable X_t. In many economic settings this setup is too restrictive because it implicitly treats X_t as exogenous and does not allow for feedback from Y_t to X_t. In the present context, such feedback is plausible (e.g., stock market movements may influence policy decisions and macroeconomic dynamics). In addition, macro-financial relationships are dynamic, and their effects may occur with lags. These features motivate a framework in which all variables are treated as endogenous and allowed to depend on lagged values of the entire system.

A useful intermediate step is the univariate autoregressive model. For example:

$$S\&P\ 500_t = \alpha + \beta S\&P\ 500_{t-1} + \varepsilon_t$$

where S&P 500 denotes the index returns at time t, and is regressed on the lagged value of itself as $S\&P\ 500_{t-1}$. A VAR extends this idea to multiple variables by allowing each variable to depend on lagged values of all variables in the system.

<u>VAR(p) specification</u>

Let y_t be an $N \times 1$ vector of endogenous variables at time t. In this thesis, y_t contains:

$$y_t = \begin{pmatrix} S\&P\ 500_t \\ M2_t \\ RGDP_t \end{pmatrix}$$

A VAR of lag-order p, denoted VAR(p)[14], is written as:

$$y_t = \alpha + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \cdots + \beta_p y_{t-p} + \varepsilon_t$$

where:

- α is an $N \times 1$ vector of intercepts,
- β_k (for $k = 1, \ldots, p$) are $N \times N$ coefficient matrices,
- ε_t is an $N \times 1$ vector of reduced-form innovations with $E[\varepsilon_t] = 0$ and $E[\varepsilon_t \varepsilon_t'] = \Sigma_\varepsilon$.

An example VAR(1) for the three-variable system is:

$$\begin{pmatrix} S\&P\ 500_t \\ M2_t \\ RGDP_t \end{pmatrix} = \alpha_n + \beta_{nk} \begin{pmatrix} S\&P\ 500_{t-1} \\ M2_{t-1} \\ RGDP_{t-1} \end{pmatrix} + \varepsilon_{nt}$$

Which can be expanded from the vectorial notation, and written equation-by-equation, the VAR(1) becomes[15]:

[14] More specifically, I am referring here to reduced-form VAR, where the residuals of the regressions are assumed to be non-correlating. In case they are correlating the residual error terms can become unreliable, which comes relevant in the IRF and FEVD subsections.

[15] Extra step for intuition in Appendix D

$$S\&P\ 500_t = \alpha_1 + \beta_{11} S\&P\ 500_{t-1} + \beta_{12} M2_{t-1} + \beta_{13} RGDP_{t-1} + \varepsilon_{1t}$$

$$M2_t = \alpha_2 + \beta_{21} S\&P\ 500_{t-1} + \beta_{22} M2_{t-1} + \beta_{23} RGDP_{t-1} + \varepsilon_{2t}$$

$$RGDP_t = \alpha_3 + \beta_{31} S\&P\ 500_{t-1} + \beta_{32} M2_{t-1} + \beta_{33} RGDP_{t-1} + \varepsilon_{3t}$$

Each equation explains one variable using lags of all variables. Importantly, VAR estimation does not designate one "main" dependent variable, rather, each variable has its own equation. Because the regressors are lagged values (predetermined) and are identical across equations, each equation can be estimated by Ordinary Least Squares (OLS), and under standard assumptions[16] this is equivalent to system estimation.

Lag length selection

Choosing the number of lags p is a key modeling decision. Too few lags may leave serial correlation in the residuals and omit important dynamics, and too many lags increase parameter uncertainty and reduce degrees of freedom. This thesis selects p using the Bayesian Information Criterion (BIC), following the recent work of Sharif et al. (2025)[17].

A convenient expression for the BIC of a VAR(p) is:

$$BIC(p) = ln\left|\hat{\Sigma}_\varepsilon(p)\right| + \frac{\ln(T)}{T} * k_p$$

Where:

- $\hat{\Sigma}_\varepsilon(p) =$ *is the estimated covariance matrix of the residuals for the VAR*(p)
- T *is the effective number of observations (after accounting for lags)*
- $k_p =$ *is the number of estimated parameters (including intercepts)*
- $ln\left|\hat{\Sigma}_\varepsilon(p)\right|$ represents the model's goodness of fit (smaller is better), while the second term penalizes the number of parameters, increasing with p
- *The optimal lag amount is the value of* p *that minimizes the BIC*

Interpretation of VAR results

Coefficient interpretation is similar to linear regression but in a multivariate and dynamic setting. A coefficient on lagged RGDP in the S&P 500 return equation (like $\beta_{13} RGDP_{t-1}$) captures whether past RGDP growth contains predictive information for current stock returns conditional on the other lagged variables in the system, which can be statistically tested via regular t-test and in this case with the 5% confidence level. Also, the residuals of the model can be used to gather information about the relationships in the model, in this case through IRF and FEVD. However, reduced-form residuals ε_t may be contemporaneously correlated across equations, so interpreting residual shocks as distinct economic shocks requires additional identification (mainly structural VAR). Since this thesis focuses on reduced-form dynamics and forecast-based interpretation, identification is handled through standard orthogonalization for IRFs/FEVD (described below).

The VAR analysis also relies on the stationarity and stability conditions of the system (already addressed via transformations and further by unit root testing). In addition, residual diagnostics (e.g., autocorrelation and heteroskedasticity checks) are used to assess robustness of inference.

[16] Appendix D (II)

[17] Other less penalizing selection criteria, like AIC (Akaike information criterion) can yield higher number of lags. BIC is chosen based on Sharif et al. (2025) approach to ensure robustness against overparameterization.

II (Granger-causality test)

Granger-causality test examines whether lagged values of one variable improve the prediction of another variable within the VAR system. In the context of the VAR(p):

$$y_t = \alpha + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \cdots + \beta_p y_{t-p} + \varepsilon_t$$

Variable x Granger-causes variable y if the lags of x enter significantly in the equation for y, conditional on lags of all other variables.

Formally, in the y-equation, the null hypothesis is:

H_0: $\beta_1 = \beta_2 = \cdots = \beta_p = 0$ (i.e. no Granger-causality)

H_A: $At\ least\ one\ \beta_k \neq 0$ (i.e. implying lagged terms Granger-cause the dependent variable)

This is implemented via an F-test. Rejection of H_0 indicates that lagged values of x provide statistically significant predictive content for y beyond what is captured by the rest of the VAR. Importantly, Granger-causality is a statement about predictability, not philosophical or structural causality, as discussed in Appendix A.

(Primary references: Granger, 1969; Granger, 1980; Lütkepohl, 2005.)

III (Impulse Response Function)

An impulse response function traces the dynamic response of the VAR system to a one-time shock. Under standard conditions, a VAR can be written in an infinite moving-average (MA) form:

$$y_t = \mu + \sum_{h=0}^{\infty} \Psi_h \, \varepsilon_{t-h}$$

where Ψ_h are matrices of dynamic multipliers and ε_t are reduced-form innovations. The element (i, j) of Ψ_h describes the effect of a one-unit reduced-form shock in equation j at time t on variable i at horizon h. This can be drawn out in a convenient graph showing the implied movement of i based on a one standard deviation shock in j, propagating through time-horizons.

Because reduced-form innovations u_t are likely contemporaneously correlated, the IRF must be orthogonalized for better interpretation. The approach chosen is standard Cholesky decomposition of the residual covariance matrix:

$$\Sigma_\varepsilon = PP'$$

where P is lower triangular. Defining orthogonal shocks ϵ_t as:

$$\varepsilon_t = P\epsilon_t, \qquad E[\epsilon_t \epsilon_t'] = I$$

the orthogonalized IRFs trace responses to shocks that are conveniently uncorrelated by construction.

A key implication is ordering sensitivity. The recursive structure imposed by Cholesky means that variables earlier in the ordering are allowed to affect variables later in the ordering contemporaneously, but not vice versa. In this thesis, ordering is chosen based on the theoretical discussion (with RGDP ordered first), and results are interpreted with the ordering dependence in mind, and robustness checks are conducted by switching the ordering.

(Main references: Sims, 1980; Lütkepohl, 2005; and applied guidance consistent with recent macro-finance VAR usage.)

IV (Forecast Error Variance Decomposition)

FEVD complements IRFs by quantifying the importance of each orthogonal shock in explaining forecast uncertainty. Specifically, for variable i, FEVD asks: what share of the h-step-ahead forecast error variance is attributable to shocks in each variable j?

Using the orthogonal MA form:

$$y_t = \mu + \sum_{j=0}^{\infty} \Theta_j \epsilon_{t-j}$$

where $\epsilon_t = P^{-1}\varepsilon_t$ are the orthogonal shocks, and $\Theta_h = \Psi_j P$ are the matrices of impulse response to these orthogonal shocks. Then the FEVD share for variable i attributable to shock j at horizon h is:

$$FEVD_{ij}(h) = \frac{\sum_{k=0}^{h-1}\left(\theta_{ij}^{(k)}\right)^2}{\sum_{m=1}^{N}\sum_{k=1}^{h-1}\left(\theta_{im}^{(k)}\right)^2}$$

For each i and h, the shares sum to 1 across all j. Thus, the share of a shock per each variable can be interpreted as a percentage of the forecast error variance. Typically, own-shocks account for the largest share, but if cross-variable dynamics are important, the contribution of other shocks grows with the horizon.

(Main references: Sims, 1980; Lütkepohl, 2005; and applied guidance consistent with recent macro-finance VAR usage.)

V (Testing for structural breaks, Bai-Perron)

Macro-financial relationships may shift due to regime changes (e.g., policy changes, major crises, institutional reforms). A structural break refers to a persistent change in the parameters of a model, such as a change in the slope coefficients of a regression, implying that the relationship between variables differs across time.

The Bai-Perron framework models multiple structural breaks by allowing coefficients to change at unknown break dates. The sample is partitioned into $m + 1$ regimes, and parameters are allowed to differ across regimes. The break dates are chosen to minimize the total Sum of Squared Residuals (SSR) across segments, and efficient algorithms search over break partitions without brute-force applications. The methodology also provides statistical procedures (sequential F-tests) to determine whether breaks exist with statistical significance and how many are supported by the data.

Two researcher-chosen inputs are required in practice:

- a maximum number of breaks m_{max},
- a trimming parameter that enforces a minimum segment length (to avoid implausibly short regimes).

In this thesis, m_{max} is set to 9[18] as a conservative upper bound given the long sample and the presence of multiple potential macro regimes, while the trimming parameter is set to 10%[19], implying a minimum regime length of roughly 26 quarters (about 6–7 years) in the quarterly dataset.

Application in this thesis:

Bai-Perron was originally developed for single-equation models, but the logic can be expanded for to multivariate systems like the VAR, following the methods displayed by Bai (2000). This method analyzes whether there are structural shifts in the parameters that significantly affect the system as a whole, not just an individual regression. This way the breaks must be more prominent and be intuitively included in most of the variables to be significant. To combat this increased detection difficulty, I will in the thesis both individually study each variable RGDPs and M2s relationship with S&P 500 for structural breaks, and the trivariate system including all variables.

The outputs of the method are (i) the number of statistically significant breaks and (ii) estimated break dates. Where break results differ across specifications, the interpretation emphasizes the multivariate specification while using the others to assess robustness.

(Main references: Bai & Perron, 1998; Bai 2000; Bai & Perron, 2003; Smith et al. 2021; Cassini & Perron, 2018)

4. Results

This section presents the empirical results obtained using the methods described in Section 3.1. Methodology to test the six hypotheses introduced in Section 2.2. Hypothesis. The hypotheses are evaluated one by one. I begin with H_V (structural breaks) because potential regime shifts affect how the remaining results should be interpreted. The remaining hypotheses are then discussed in the order presented in Section 2.2.

Results are presented in a summary form that highlights the main findings while avoiding unnecessary clutter. Detailed tables, figures, and full output snapshots are provided in the referenced appendices, together with a link to the repository containing replication code and extended results Appendix E (I). Interpretations are kept brief in this chapter and are consolidated in the Conclusions section.

4.1. Results

I (H_V: Data has structural breaks for both variables)

I begin by testing for multiple structural breaks using the Bai-Perron methodology. The goal is to assess whether the estimated relationships appear stable or whether there are statistically supported regime shifts that would motivate splitting the sample.

[18] The number 9 is chosen based on the amount of FRED (2025B) recognized U.S. recessions in the timeframe, and it serves as a sufficiently large number that it should never be reached by conventional breaks

[19] This is based on choosing the largest trimming parameter that can still assign a distinct regime for the gap between dotcom and GFC.

RGDP–S&P 500 specification

Applying the break test to the RGDP–S&P 500 specification yields no statistically significant structural breaks in the quarterly dataset, the quarterly outlier-removed dataset, or the yearly dataset. When the number of breaks is mechanically forced, the procedure often places a breakpoint around 1977–1978, which is economically plausible (stagflation-era regime change), but these breakpoints are not supported by the formal significance tests and are therefore treated as suggestive rather than conclusive.

M2–S&P 500 specification

Repeating the same process for the M2–S&P 500 specification again yields no statistically significant breaks across the quarterly, outlier-removed quarterly, and yearly samples. Under forced-break specifications, estimated break dates vary more, with some clustering around the early-2000s to GFC period, but without statistical support.

Trivariate VAR (S&P 500, RGDP, M2)

Finally, applying break testing to the trivariate setting also yields no statistically significant evidence of breaks in the baseline specifications. When forcing breaks, dates around 2008–2009 sometimes appear in the return equation, and 2018 appears more often in equations where S&P 500 predicts RGDP/M2; however, the latter pattern weakens notably in the outlier-removed dataset.

Conclusion

Overall, the Bai–Perron procedure does not provide statistically significant evidence of structural breaks under the chosen settings. Accordingly, the baseline analysis in the remaining hypotheses uses the full-sample estimates. At the same time, because structural break tests can have limited power in noisy macro-financial settings and under trimming constraints, I also report informal subsample checks around the recurring (but statistically unsupported) candidate dates, approximately 1977–1978, 2002–2009, and 2018, as supplementary perspective rather than as primary identification.

II (H_I & H_{II}: Real economic & liquidity channels exist)

This subsection evaluates whether lagged RGDP growth and lagged M2 growth exhibit predictive content for S&P 500 returns (the 'real activity' and 'liquidity' channels, respectively). I assess these hypotheses using VAR coefficient significance and Granger-causality tests as primary evidence, with IRFs and FEVD as supporting tools (Appendix E contains the full set of output).

H_I(Real activity channel: RGDP → S&P 500)

- **Quarterly baseline**: No statistically significant evidence is found in the quarterly full-sample estimates.
- **Quarterly outlier-removed**: Results improve slightly; one lag of RGDP occasionally approaches significance (often surpassing the 10% level), but not consistently at the 5% level. Subsample checks suggest the relationship is weaker in later periods.
- **Yearly dataset**: Evidence becomes statistically significant: the first lag of RGDP in the S&P 500 equation is significant at the 5% level, and the Granger-causality test rejects the null of no Granger-causality at the 5% level (Appendix E (II)). The estimated sign is negative, contrary to the baseline expectation.

Taken together, the evidence suggests that any RGDP → return relationship is horizon-dependent: it is not detectable at quarterly frequency in a stable way, but becomes detectable in annual data. The sign and time-variation imply that the 'channel' is not straightforwardly stable or monotone across regimes.

H_{II} (Liquidity channel: M2 → S&P 500)

- **Quarterly baseline and outlier-removed**: No statistically significant evidence is detected, and subsample results do not reveal a stable pattern.
- **Yearly dataset**: Statistically significant results appear, with the strongest effect typically on the second lag and a negative estimated sign. Granger-causality tests support predictive content at the 5% level (Appendix E (III)). The relationship again appears stronger earlier in the sample and weaker later.

Conclusions

Evidence consistent with both channels is found primarily in the annual data. However, the effects are negative and appear time-varying, which deviates from the simplest theoretical interpretation and suggests the relationships may be dominated by discount-rate dynamics, regime dependence, or omitted factors rather than stable structural effects.

III (H_{III} & H_{IV}: Anticipation channels exist)

This subsection evaluates whether stock market returns contain predictive content for subsequent RGDP growth and M2 growth (the 'anticipation' channels). Operationally, this is assessed by examining the RGDP equation and the M2 equation in the relevant VAR systems and testing whether lagged S&P 500 returns are jointly significant (Granger-causality).

H_{III} (S&P 500 → RGDP)

- **Quarterly baseline**: The first lag of S&P 500 returns in the RGDP equation is statistically significant with a small positive estimate, and the Granger causality test rejects the null of no predictability. The effect is not strongly dependent on subsample splits; changes mostly shift which lag appears most relevant, consistent with timing differences across regimes.
- **Quarterly outlier-removed**: The effect weakens somewhat, especially in later periods, but the results remain broadly consistent with predictability.
- **Yearly dataset**: Predictability largely disappears; lagged S&P 500 coefficients become insignificant, and the Granger test fails to reject the null (Appendix E (IV)).

Overall, the evidence supports H_{III} as a short-horizon phenomenon visible at quarterly frequency, consistent with markets embedding near-term expectations, while not as an annual relationship.

H_{IV} (S&P 500 → M2)

- **Quarterly baseline**: Full-sample estimates do not show statistically significant predictability from S&P 500 returns to M2 growth.
- **Subsamples**: Some shorter-window results show episodic significance, but they are not robust across specifications.
- **Outlier-removed and yearly**: Evidence weakens further, suggesting that any apparent predictability is sensitive and may be driven by extreme episodes (notably around Covid-era dynamics). Appendix E (V) summarizes the main outputs.

Conclusions

I find relatively strong evidence supporting H_{III} (returns predict RGDP) at quarterly frequency, but this effect does not persist in annual data. For H_{IV} (returns predict M2), robust evidence is not found; any significance appears episodic and specification dependent.

IV (H_{VI}: Comparing both variables (RGDP vs M2))

This subsection compares the predictive content of RGDP and M2 for S&P 500 returns within the joint trivariate VAR. I compare coefficient estimates and significance in the S&P 500 equation, Granger-causality outcomes, IRFs, and FEVD shares.

Quarterly joint VAR

Consistent with earlier results, quarterly estimates show minimal predictive content from either RGDP or M2 for S&P 500 returns: coefficients are small relative to standard errors and lack statistical significance. Granger causality, IRFs, and FEVD similarly do not support meaningful explanatory power at quarterly horizons. Across specifications, the more robust dynamics are instead found among the non-return equations (e.g., S&P 500 → RGDP, M2 → RGDP, RGDP → M2), as summarized in Appendix E (VI). The results are effectively the same for the outlier-removed set.

Yearly joint VAR

In annual data, statistical significance reappears, but it is concentrated on the RGDP term: RGDP exhibits significant predictive content for returns (with a negative estimate around -2%), while M2 remains insignificant in the return equation. The IRFs show visibly stronger responses of S&P 500 to RGDP shocks than to M2 shocks, and FEVD indicates a larger variance share attributable to RGDP than to M2 (Appendix E (VII)). Subsample checks suggest that even the RGDP relationship weakens in the latter part of the sample, aligning with the broader theme of instability.

Stepping aside from the chosen lag selection criteria suggesting $p = 1$, forcing lags at $p = 2$, yields different results, with M2 gaining explanatory power, while RGDP seems to lose it. This results is illustrated in Appendix E (VIII), and shows signs that the comparative results are highly time scale sensitive.

Conclusions

At quarterly frequency, both variables are similarly weak predictors of S&P 500 returns. At annual frequency, RGDP shows clearer explanatory importance than M2 across coefficients, IRFs, FEVD, and Granger-causality results, although the relationship appears weaker in later periods.

4.2. Robustness of results

This section discusses the main factors affecting robustness and explains how they influence interpretation. The key limitations concern: (i) non-normality and heavy tails, (ii) finite-sample limitations (especially in annual data), (iii) specification sensitivity (lag length, ordering, subsamples, outliers), and (iv) heteroskedasticity/autocorrelation and standard error choices.

I (Distributional assumptions and non-normality)

Many inference procedures used in this thesis, such as t-tests for individual coefficients and F-tests for joint restrictions (including Granger-causality), are derived under normality assumptions for small-sample results. In macro-financial time series, non-normality is common due to skewness, fat tails, volatility clustering, and crisis episodes.

To assess this empirically, residual normality was tested using the Jarque-Bera (JB) test[20], which null hypothesis is normally distributed errors. The JB test strongly rejects normality across datasets[21], indicating that reported p-values should be interpreted as approximate rather than exact.

A second practical point is that JB-type tests are highly sensitive to outliers; a small number of extreme observations can drive rejection mechanically. This matters here because the Covid-era episode produces unusually large moves in both macro and financial variables. For this reason, the thesis includes an outlier-removed quarterly dataset and compares results across inclusive vs. exclusive samples as a partial robustness check.

Importantly, non-normality does not automatically invalidate VAR estimation itself: OLS estimation remains consistent under standard weak conditions, in line with standard practice (Cassini et al. 2018). The main concern is finite sample inference (how reliable the reported significance tests are). For IRFs the use of bootstrapped confidence intervals partially mitigates distributional concerns by avoiding reliance on normal approximations for those specific outputs.

II (Sample size, frequency choice, and statistical power)

A central robustness constraint is finite sample size and the implied power of tests. The quarterly dataset contains 261 observations, while the yearly dataset contains only 64 observations. The annual sample is therefore especially vulnerable to:

- Reduced degrees of freedom after including lags.
- Higher parameter uncertainty.
- Sensitivity to individual episodes and regime changes.

At the same time, annual aggregation can increase detectability of slow-moving relationships by reducing high-frequency noise (which may explain why some predictive relationships appear in annual data but not quarterly data). This creates an inherent trade-off: quarterly data offers more observations but may obscure low-frequency relationships, while yearly data aligns better with long-horizon dynamics but is statistically fragile. Accordingly, this thesis emphasizes patterns that replicate across multiple diagnostics (coefficients + Granger + IRF/FEVD) rather than relying on a single test.

III (Specification sensitivity; lags, ordering, subsamples, outliers)

Several robustness points arise from model specification choices:

- **Lag length selection**: Results can change with p. The thesis uses BIC to select lags, but finite-sample sensitivity remains possible. Where results are borderline, interpretation is cautious and cross-checked across specifications.
- **Cholesky ordering for IRFs/FEVD**: Orthogonalized IRFs and FEVD depend on the ordering assumption. This thesis uses an ordering guided by theory (RGDP first) but interprets IRF/FEVD evidence qualitatively, acknowledging that alternative orderings could shift the magnitude and timing of responses. Again, results are cross-checked by switching the ordering.
- **Subsample splits**: The thesis reports exploratory splits around candidate dates suggested (weakly) by break-testing and historical context. However, repeated subsample searching increases the risk of data-mining. Therefore, subsample results are treated as descriptive rather than as primary evidence.

[20] Jarque & Bera (1980)
[21] All exact diagnostics can be found at Appendix E (I)

- **Outliers**: Covid-era outliers meaningfully affect results in some specifications, especially for predictability claims involving M2. The outlier-removed dataset serves as a direct check on whether conclusions are driven by a small number of extreme observations.

IV (Heteroskedasticity, autocorrelation, and standard errors)

Coefficient inference is based on conventional (non-HAC) standard errors, which can be problematic if residuals are heteroskedastic or autocorrelated. To evaluate this, residuals were tested using ARCH tests (conditional heteroskedasticity) and the Portmanteau test (serial correlation). The yearly dataset shows no strong evidence of ARCH effects or autocorrelation, while the quarterly dataset shows some indications of heteroskedasticity but limited autocorrelation[22].

Because the thesis's most consequential significance findings arise mainly in the yearly dataset—and diagnostics there do not indicate strong violations—the use of conventional standard errors is unlikely to overturn the headline conclusions. Nevertheless, for borderline quarterly results, inference is interpreted conservatively, and robustness is assessed through cross-sample comparisons (outlier removal, subsamples, and annual aggregation) rather than relying solely on nominal p-values.

5. Conclusions

This thesis examined the dynamic relationships between U.S. stock market returns, real economic activity, and monetary expansion (liquidity conditions) using reduced-form VAR methods. Based on the literature review, I expected to find evidence that both real economic growth and monetary expansion contain predictive information for stock market returns, primarily over longer horizons, and that stock market returns may also predict subsequent movements in real activity and liquidity conditions (anticipation channels). I further aimed to compare the relative explanatory importance of real growth versus monetary expansion for stock returns, a topic for which the existing empirical evidence remains limited and often sensitive to specification choices.

Overall, the results provide selective support for these expectations: relationships are most visible in the annual data (for the macro → returns direction), while the quarterly data mainly supports a short-horizon anticipation effect from returns to real activity, and real economic growth shows greater explanatory power to monetary expansion. At the same time, several findings deviate from baseline theoretical expectations, most notably the negative sign of macro predictors in the annual return equation and the weak evidence for returns predicting M2 outside specific episodes.

I (Structural breaks)

The literature review suggested that macro-financial relationships and return predictability may be unstable over time, motivating an expectation of structural breaks, particularly around major regime shifts and the turn of the millennium. I tested for multiple breaks using the Bai-Perron methodology. (Smith et al. 2023; Bai, 2003)

[22] References for the tests; ARCH: Engle (1982), Portmanteau: Box & Pierce (1970)

Across the specifications tested, I found no statistically significant structural breaks. Break dates appeared only when the model was forced to include breaks, with recurring (but statistically unsupported) candidates around 1977–1978, parts of the early-2000s to GFC window, and occasionally 2008–2009.

These results should be interpreted carefully. A 'no-break' result means that there is no statistically detectable evidence of abrupt parameter shifts. Bai-Perron tests are well-suited to identifying discrete breakpoints, but they can be less sensitive to gradual drift or to instability that manifests primarily in volatility and higher moments rather than in conditional means. In practical terms, based on this, the thesis proceeded primarily with full-sample inference, using subsamples only as descriptive checks rather than as the central identification strategy. (Bai, 2003)

II (Real activity and liquidity channels)

Consistent with the literature, I expected real growth and monetary expansion to be positively associated with stock returns over longer horizons, while not so much for the shorter term (Fichtner et al. 2024). In the data, the results are horizon-dependent, but the signs are negative, not as expected.

- **Quarterly frequency**: Neither RGDP nor M2 shows robust predictive power for stock returns in the quarterly VARs. This aligns with the view that high-frequency returns contain substantial noise and are dominated by fast-moving financial components rather than slow macro aggregates.
- **Annual frequency**: Evidence of predictability appears for both RGDP and M2 in bivariate settings, but the estimated relationships are negative and appear weaker in later subsamples.

The annual-frequency results support the broader literature theme that macro–stock relationships can be more detectable over longer horizons, though the instability across periods is consistent with recent findings that the stock–growth relationship can be fragile and sample-dependent. (Fama, 1990; Rigobon et al. 2003)

<u>Interpreting the negative sign</u>

The negative sign of RGDP/M2 in the annual return equation is a central puzzle relative to the simplest theoretical story. Several mechanisms can rationalize this within a reduced-form framework:

- **Policy reaction and countercyclicality**: Liquidity expansion often occurs in weak economic or market conditions (e.g., easing during downturns). In reduced-form regressions, this can produce a negative association between monetary expansion and contemporaneous/near-term returns even if the structural effect of easier liquidity on valuations is positive.
- **Discount-rate dominance**: Stock prices and returns respond not only to expected cash flows but also to changes in discount rates. If stronger growth (or growth expectations) coincides with higher expected returns (higher discount rates), prices can fall even when cash-flow prospects improve, generating negative return relations in reduced-form estimates.
- **Measurement and timing mismatch**: RGDP and M2 are slow-moving, revised, and observed at low frequency, while equity markets are forward-looking. Reduced-form regressions using lagged macro aggregates can therefore capture delayed or policy-mediated components rather than the 'clean' theoretical channel.

Given these factors, the results support the existence of a long-horizon relationship in the data, but they also suggest that the relationship is not a simple positive structural mapping from macro improvements to higher equity returns.

III (Anticipation channels)

The anticipation channel hypotheses concerned whether stock returns predict subsequent movements in real activity and liquidity conditions. Here, the results are more aligned with standard intuition for the real economy but not for monetary expansion.

- **Returns → RGDP**: I find evidence that S&P 500 returns predict RGDP growth at quarterly horizons, but this relationship does not persist in annual data. This is consistent with the idea that equity returns embed forward-looking information about economic conditions, and that the market can lead measured real activity over short horizons. This is broadly compatible with work linking stock returns to real activity expectations and macro news. (Fama, 1981)
- **Returns → M2**: I do not find robust evidence that stock returns predict M2 growth outside specific, crisis-driven episodes. A likely explanation is that M2 is a broad monetary aggregate that adjusts smoothly and is not a clean high-frequency measure of discretionary policy shocks (relative to, for example, policy rates or high-frequency monetary surprises). Interpreted conservatively, the results suggest that broad monetary expansion, as measured by M2, does not strongly or reliably respond to stock returns in a way that is detectable in this reduced-form setup.

IV (Comparing real growth and monetary expansion)

The central comparative question of the thesis is whether real growth (RGDP) or monetary expansion (M2) provides greater explanatory importance for stock returns when both are included jointly.

- **Quarterly joint model**: Both variables are weak predictors of stock returns; significance is largely absent, consistent with the earlier quarterly bivariate results.
- **Annual joint model**: RGDP shows clearer dominance over M2. In the joint system, RGDP retains predictive content for returns while M2's predictive content weakens substantially. IRF and FEVD diagnostics likewise suggest a larger role for RGDP shocks than for M2 shocks in explaining longer-horizon return variation. Furthermore, RGDP shocks appear to propagate faster than those of M2.

Two interpretations are particularly consistent with this outcome:

- **Shared information and collinearity**: If M2 and RGDP are correlated[23], then bivariate predictability from M2 may partially reflect information that is more directly captured by RGDP in the joint model. In that case, once RGDP is included, M2 adds little incremental predictive content.
- **Different propagation horizons**: The IRF evidence suggests that the timing of responses differs across shocks. If M2 effects propagate more slowly than RGDP (or are mediated through real activity), lag length and frequency choice can materially affect which variable appears dominant. This is consistent with the observation that alternative lag choices can shift which predictor appears significant, even if such alternatives are not preferred by the BIC.

Bottom line. Within the chosen reduced-form framework and sample: real economic growth appears more informative than broad money growth for long-horizon stock return predictability, while both are weak at quarterly horizons. At the same time, the weakening of effects in later subsamples and the lack of statistically significant breakpoints suggest that instability may be present in a gradual or low-power sense rather than as sharp regime shifts.

[23] See the quantity theory of money formula in the literature review, by Friedman (1987).

V (Practical implications)

This thesis does not provide a mechanical strategy for forecasting equity markets. However, the results yield several practical implications for how different actors may interpret macro-financial information.

- **Horizon matters more than 'which variable'**: A consistent takeaway is that macro proxies (RGDP and M2) have little detectable predictive content for stock returns at quarterly horizons, while some relationships become visible at annual horizons. Practically, this suggests that short-term equity return variation is likely driven primarily by factors not captured by broad macro aggregates (e.g., discount-rate news, risk premia variation, sentiment, or market-specific shocks), whereas macro conditions may be more useful when evaluating long-run market environments rather than near-term return moves.
- **Real activity signals appear more informative than broad money growth in joint settings**: In the annual joint model, RGDP dominates M2 in explaining return dynamics (in coefficients, Granger evidence, and variance decomposition). For decision-makers who must prioritize a limited set of macro indicators (e.g., strategic asset allocation committees, risk managers, macro analysts, or even individual investors), this implies that real activity measures may provide more incremental information about the market's medium-to-long horizon environment than broad money growth, at least within the sample and framework used here.
- **Liquidity aggregates may reflect policy reaction rather than policy impact**: The negative sign and weak incremental contribution of M2 in the joint annual model are consistent with the idea that broad money growth can be endogenously rising in response to weak conditions and stress. Practically, this implies that interpreting money growth mechanically as 'fuel for equities' can be misleading unless the analysis separates policy reaction from policy shocks. For practitioners, this supports using M2 as a descriptive indicator of the liquidity regime but treating its relationship with returns as conditional on broader macro context.
- **Stock markets can lead the real economy at short horizons**: The evidence that quarterly stock returns predict RGDP growth supports the view that markets embed forward-looking information and may function as an early signal of near-term changes in real activity. For business-cycle monitoring, scenario planning, or macro risk analysis, equity returns may therefore be informative as a high-frequency indicator, while recognizing that this is about statistical predictability, not guaranteed causality.
- **Policy and investment decisions should not rely on small macro-return coefficients**: Even where statistically significant relationships appear, the estimated effects are small and appear unstable across time. Practically, this means the thesis supports interpretation and contextual understanding more than point forecasting. For instance, it is more defensible to use the findings to shape expectations about how macro regimes relate to returns over years, rather than to infer specific quarterly return predictions from macro releases.

Taken together, even when predictive power is limited, systematically comparing real activity and monetary expansion clarifies what kind of macro information is useful, at what horizons, and with what caveats, valuable for both applied finance and macroeconomic interpretation.

VI (Limitations and avenues for further research)

Several limitations also motivate clear extensions:

- **Richer measures of liquidity and monetary policy shocks**: M2 is a broad aggregate and may not capture discretionary policy shocks or liquidity conditions as directly as policy rates, high-frequency monetary surprises, credit spreads, or financial condition indices. Future work could use alternative liquidity proxies, while explicitly discussing comparability with real activity measures.
- **Structural identification**: This thesis uses reduced-form VAR tools (including Cholesky-orthogonalized IRFs/FEVD). Stronger causal interpretation would require structural identification (e.g.,

structural VAR restrictions or external instruments) to separate policy shocks from endogenous responses.

- **Alternative horizons and mixed-frequency approaches**: Quarterly/annual aggregation may hide shorter-term transmission mechanisms. Mixed-frequency methods or higher-frequency financial variables combined with quarterly macro data could help disentangle timing more precisely.
- **Instability beyond mean breaks**: Bai-Perron targets breaks in conditional-mean parameters. Instability in volatility, tail behavior, or time-varying parameters could exist even when mean-break tests are insignificant. Models with time-varying parameters or regime-switching volatility could therefore provide a more complete picture.
- **More comprehensive measure of the stock market**: While the S&P 500 provides a simple and accessible variable to proxy the U.S stock market, another more comprehensive portfolio could allow to detail the effects even better. While dividing the market into industry portfolios could be seen as a logical next step to identify clearer causal patterns in the relationships.

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Appendix A

The Granger Causality Test and Its Criticisms

The Granger-causality test, introduced by Granger (1969), is one of the most widely used econometric tools for assessing directional predictability between time-series variables. Under Granger's definition, a variable X 'Granger-causes' Y if past values of X contain unique information that helps predict Y beyond the information contained in past values of Y alone. The test is typically implemented in a vector autoregression (VAR) framework, comparing restricted and unrestricted models to determine whether lagged terms of X significantly improve the forecast of Y.

Formally, X is said to Granger-cause Y if:

$$\mathrm{Var}\,(Y_t \mid Y_{t-1}, \dots, Y_{t-p}) > \mathrm{Var}\,(Y_t \mid Y_{t-1}, \dots, Y_{t-p}, X_{t-1}, \dots, X_{t-p})$$

where a reduction in forecast error variance implies predictive content. This approach is grounded in the idea that causality is fundamentally a matter of predictability, not philosophical causality (Granger, 1980).

Despite its widespread adoption, the Granger test has faced substantial criticism. One major critique concerns non-causal correlations induced by omitted variables, measurement error, or common shocks. Sims (1972) famously argued that Granger-causality tests may detect spurious causality if relevant variables are excluded from the system, an issue sometimes referred to as "omitted variable bias in temporal precedence tests." Another key criticism is non-stationarity, as shown by Engle and Granger (1987), who demonstrated that standard Granger tests fail under cointegration unless error-correction terms are included. In such cases, failing to account for long-run equilibrium relationships leads to misleading inference about short-run causality.

A further critique involves structural breaks and parameter instability, which can invalidate the test's conclusions. Bai et al. (1998, 2003) and Perron (2005) show that when the data-generating process experiences regime changes, traditional Granger tests, assuming constant parameters, may falsely detect or miss causality. This is especially problematic in macroeconomic or financial time series where structural change is common.

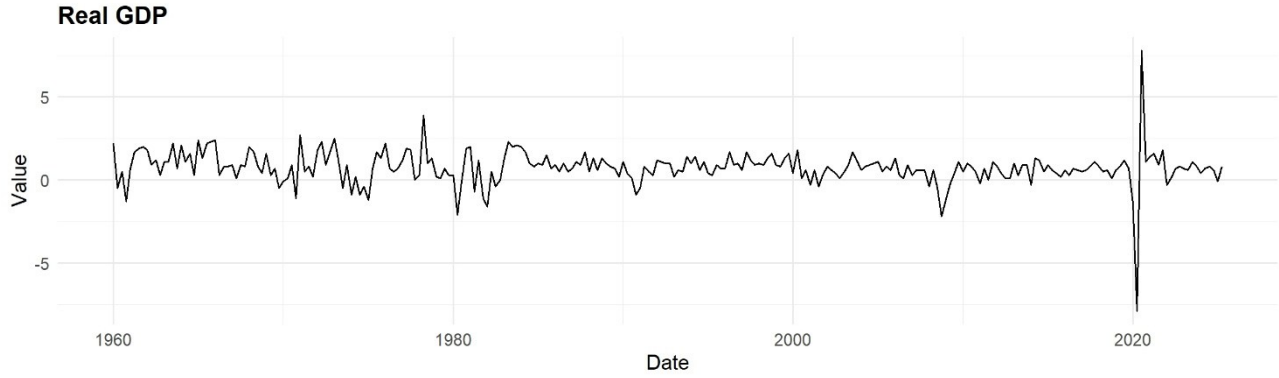
Another limitation is the test's inability to distinguish true causality from "information-based predictability." As noted by Lütkepohl (2005), the Granger framework only shows whether X contains predictive information about Y, not whether X has a structural or economic effect on Y. This distinction is central in modern causal inference, where researchers emphasize identification strategies based on exogenous variation rather than purely statistical predictability.

Finally, Granger causality is limited in the presence of nonlinear dynamics, leading to tests with low power against nonlinear alternatives. Diks & Panchenko (2006) propose nonlinear Granger-causality tests, but these are considerably more complex and still subject to small-sample distortions.

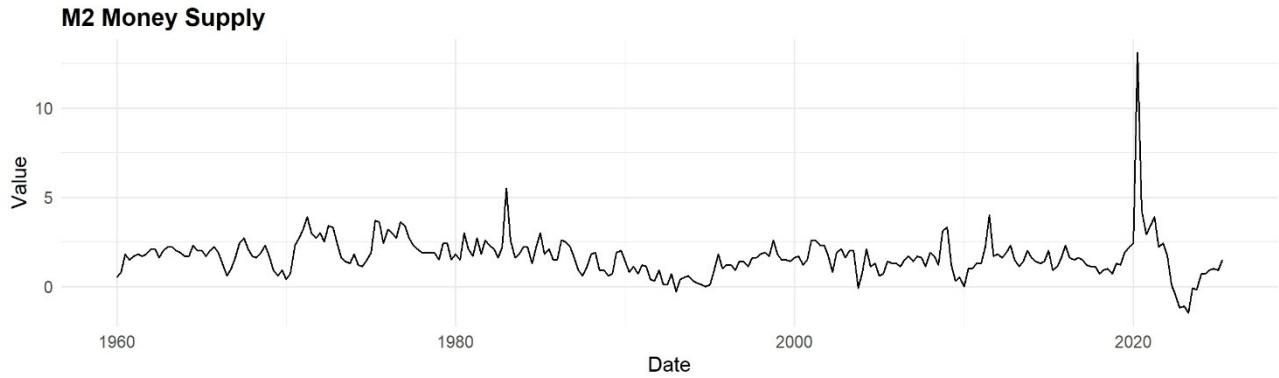
In summary, while the Granger-causality test remains foundational for empirical time-series analysis and continues to be applied across macroeconomics, finance, and data science, it should be viewed as a test of predictive precedence, not structural causality. In the case of this thesis, it is important to distinguish that results from this test don't in any way prove philosophical causality between objects but rather provided evidence for a certain predictive power in their relationship, although this is dependent on the research method accounting for non-stationarity, structural breaks, omitted variables and potential nonlinearities.

Appendix B

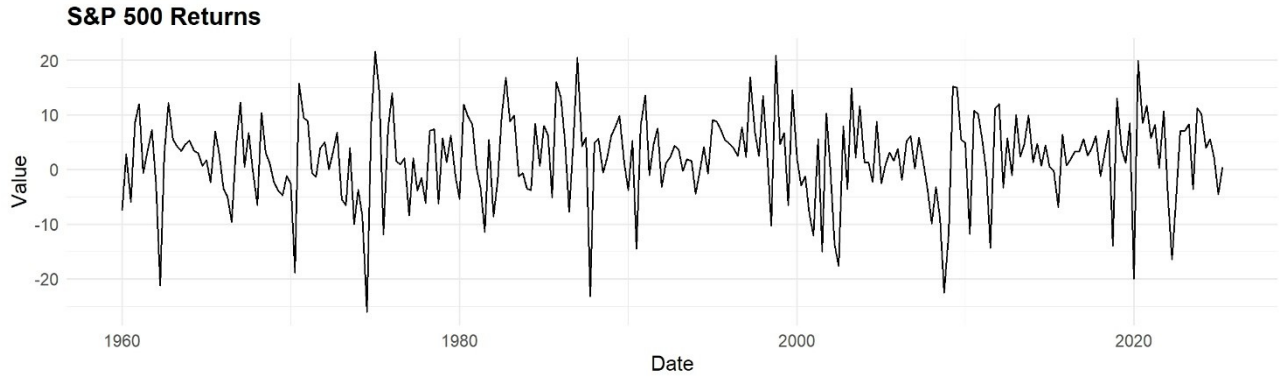
Graph I



Graph II

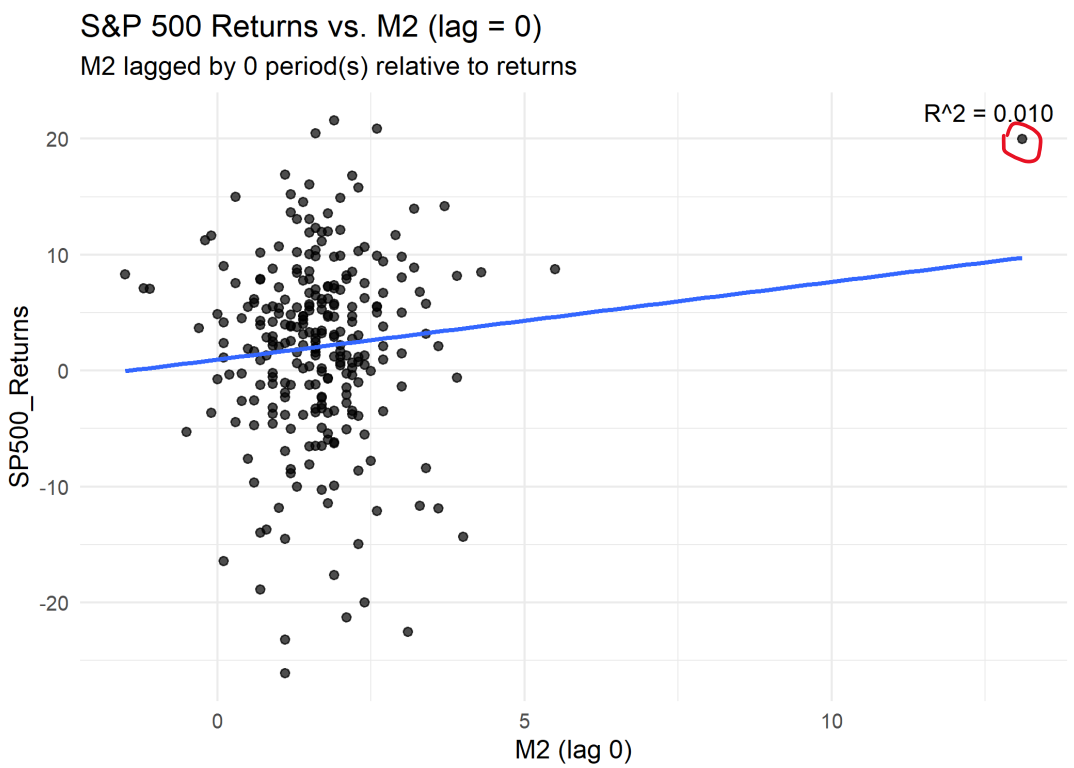


Graph III



Graph IV

Graph V



Appendix C

```
====================================
Variable: S&P 500 Returns
====================================
                    Test Statistic CV_1pct CV_5pct CV_10pct          Decision_5pct
      ADF (ur.df, trend)  -11.8047 -3.9800 -3.4200  -3.1300        Reject H0 at 5%
       PP (Z-tau, trend)  -15.2817 -3.9965 -3.4283  -3.1373        Reject H0 at 5%
              KPSS (tau)    0.0490  0.2160  0.1460   0.1190 Fail to reject H0 at 5%
 ADF-GLS (DF-GLS, trend)   -4.9154 -3.4800 -2.8900  -2.5700        Reject H0 at 5%
                    Null_H0
 Unit root (non-stationary)
 Unit root (non-stationary)
           Trend-stationary
 Unit root (non-stationary)

--- Quick interpretation (5% level) ---
Most unit-root tests reject H0 and KPSS does not reject stationarity → evidence points towards stat
ionarity (around a trend).

====================================
Variable: Real GDP
====================================
                    Test Statistic CV_1pct CV_5pct CV_10pct          Decision_5pct
      ADF (ur.df, trend)  -10.3825 -3.9800 -3.4200  -3.1300        Reject H0 at 5%
       PP (Z-tau, trend)  -15.9228 -3.9965 -3.4283  -3.1373        Reject H0 at 5%
              KPSS (tau)    0.0369  0.2160  0.1460   0.1190 Fail to reject H0 at 5%
 ADF-GLS (DF-GLS, trend)   -5.1715 -3.4800 -2.8900  -2.5700        Reject H0 at 5%
                    Null_H0
 Unit root (non-stationary)
 Unit root (non-stationary)
           Trend-stationary
 Unit root (non-stationary)

--- Quick interpretation (5% level) ---
Most unit-root tests reject H0 and KPSS does not reject stationarity → evidence points towards stat
ionarity (around a trend).

====================================
Variable: M2
====================================
                    Test Statistic CV_1pct CV_5pct CV_10pct          Decision_5pct
      ADF (ur.df, trend)   -6.9985 -3.9800 -3.4200  -3.1300        Reject H0 at 5%
       PP (Z-tau, trend)   -9.0789 -3.9965 -3.4283  -3.1373        Reject H0 at 5%
              KPSS (tau)    0.1036  0.2160  0.1460   0.1190 Fail to reject H0 at 5%
 ADF-GLS (DF-GLS, trend)   -3.9764 -3.4800 -2.8900  -2.5700        Reject H0 at 5%
                    Null_H0
 Unit root (non-stationary)
 Unit root (non-stationary)
           Trend-stationary
 Unit root (non-stationary)

--- Quick interpretation (5% level) ---
Most unit-root tests reject H0 and KPSS does not reject stationarity → evidence points towards stat
ionarity (around a trend).
```

Appendix D

I (VAR(1) – (S&P 500, M2, RGDP))

$$\begin{pmatrix} S\&P\ 500_t \\ M2_t \\ RGDP_t \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} + \begin{pmatrix} \beta_{11}\beta_{12}\beta_{13} \\ \beta_{21}\beta_{22}\beta_{23} \\ \beta_{31}\beta_{32}\beta_{33} \end{pmatrix} \begin{pmatrix} S\&P\ 500_{t-1} \\ M2_{t-1} \\ RGDP_{t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \\ \varepsilon_{3t} \end{pmatrix}$$

II (Standard assumptions of reduced form VAR)

- Conditional mean of y_t is adequately captured by linear function
- Innovations are white noise (zero-mean) with well-defined covariances
- Stability in stationarity transforms
- Current innovations are orthogonal to the information in past y, $E[\,\varepsilon_t \mid y_{t-1}, y_{t-2}, \dots\,] = 0$
- Time-invariant parameters, no structural changes
- No perfect multicollinearity in regressors

(Lütkepohl, 2005)

Appendix E

I (GitHub)

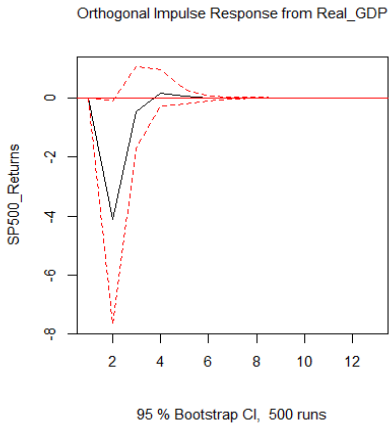
GitHub repository for all the detailed results, and the R script used for running the VAR & Bai-Perron:

https://github.com/vilpertn/Thesis-2025/tree/main

II (H_I with yearly data)

```
Estimation results for equation SP500_Returns:
==============================================
SP500_Returns = SP500_Returns.l1 + Real_GDP.l1 + const

                   Estimate Std. Error t value Pr(>|t|)
SP500_Returns.l1 -0.009675   0.129829  -0.075    0.941
Real_GDP.l1      -2.120572   0.977453  -2.169    0.034 *
const            15.258323   3.421434   4.460 3.59e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Residual standard error: 15.91 on 61 degrees of freedom
Multiple R-Squared: 0.08077,    Adjusted R-squared: 0.05063
F-statistic:  2.68 on 2 and 61 DF,  p-value: 0.07664
```

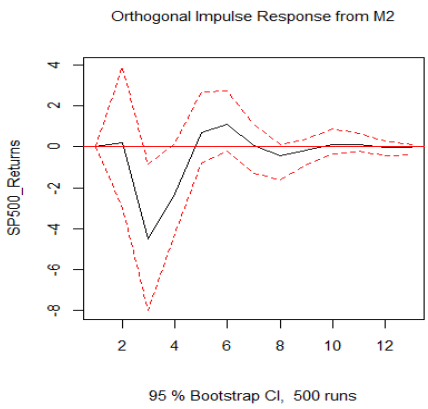
```
--- Does Real_GDP Granger-cause SP500_Returns? ---
$Granger

        Granger causality H0: Real_GDP do not Granger-cause SP500_Returns

data:  VAR object res$model
F-Test = 4.7067, df1 = 1, df2 = 122, p-value = 0.03198
```



III (H_{II} with yearly data)

```
Estimation results for equation SP500_Returns:
==============================================
SP500_Returns = SP500_Returns.l1 + M2.l1 + SP500_Returns.l2 + M2.l2 + const

                 Estimate Std. Error t value Pr(>|t|)
SP500_Returns.l1 -0.14727    0.12669  -1.162 0.249820
M2.l1             0.07254    0.62361   0.116 0.907799
SP500_Returns.l2 -0.11926    0.12563  -0.949 0.346389
M2.l2            -1.51668    0.66687  -2.274 0.026662 *
const            20.76623    5.12048   4.056 0.000151 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Residual standard error: 15.78 on 58 degrees of freedom
Multiple R-Squared: 0.1301,     Adjusted R-squared: 0.07013
F-statistic: 2.169 on 4 and 58 DF,  p-value: 0.08377
```

```
--- Does M2 Granger-cause SP500_Returns? ---
$Granger

         Granger causality H0: M2 do not Granger-cause SP500_Returns

data:  VAR object res$model
F-Test = 3.1066, df1 = 2, df2 = 116, p-value = 0.0485
```



IV (H_{III} with yearly data in the left picture and quarterly data in the right picture)

```
Estimation results for equation Real_GDP:
=========================================
Real_GDP = SP500_Returns.l1 + Real_GDP.l1 + const

                 Estimate Std. Error t value Pr(>|t|)
SP500_Returns.l1  0.04219    0.02867   1.471 0.151589
Real_GDP.l1       0.11901    0.17391   0.684 0.499010
const             2.75798    0.72870   3.785 0.000687 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Residual standard error: 2.417 on 30 degrees of freedom
Multiple R-Squared: 0.09298,    Adjusted R-squared: 0.03251
F-statistic: 1.538 on 2 and 30 DF,  p-value: 0.2313
```

```
Estimation results for equation Real_GDP:
=========================================
Real_GDP = SP500_Returns.l1 + Real_GDP.l1 + const

                 Estimate Std. Error t value Pr(>|t|)
SP500_Returns.l1 0.038082   0.007954   4.788 2.85e-06 ***
Real_GDP.l1      0.018412   0.059672   0.309    0.758
const            0.645291   0.078762   8.193 1.21e-14 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Residual standard error: 1.021 on 257 degrees of freedom
Multiple R-Squared: 0.08289,    Adjusted R-squared: 0.07575
F-statistic: 11.61 on 2 and 257 DF,  p-value: 1.483e-05
```

```
--- Does SP500_Returns Granger-cause Real_GDP ? (bivariate VAR) ---
$Granger

        Granger causality H0: SP500_Returns do not Granger-cause Real_GDP

data:  VAR object subvar
F-Test = 2.1651, df1 = 1, df2 = 60, p-value = 0.1464
```

```
--- Does SP500_Returns Granger-cause Real_GDP ? (bivariate VAR) ---
$Granger

        Granger causality H0: SP500_Returns do not Granger-cause Real_GDP

data:  VAR object subvar
F-Test = 22.921, df1 = 1, df2 = 514, p-value = 2.211e-06
```

V (H_{IV} with yearly data in the left picture and quarterly data in the right picture)

```
Estimation results for equation SP500_Returns:
==============================================
SP500_Returns = SP500_Returns.l1 + M2.l1 + SP500_Returns.l2 + M2.l2 + const

                 Estimate Std. Error t value Pr(>|t|)
SP500_Returns.l1 -0.14727    0.12669  -1.162 0.249820
M2.l1             0.07254    0.62361   0.116 0.907799
SP500_Returns.l2 -0.11926    0.12563  -0.949 0.346389
M2.l2            -1.51668    0.66687  -2.274 0.026662 *
const            20.76623    5.12048   4.056 0.000151 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Residual standard error: 15.78 on 58 degrees of freedom
Multiple R-Squared: 0.1301,     Adjusted R-squared: 0.07013
F-statistic: 2.169 on 4 and 58 DF,  p-value: 0.08377
```

```
Estimation results for equation M2:
===================================
M2 = SP500_Returns.l1 + M2.l1 + const

                  Estimate Std. Error t value Pr(>|t|)
SP500_Returns.l1 -0.010741   0.007592  -1.415    0.158
M2.l1             0.559851   0.051937  10.779  < 2e-16 ***
const             0.758352   0.105458   7.191 7.01e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Residual standard error: 0.971 on 257 degrees of freedom
Multiple R-Squared: 0.3116,     Adjusted R-squared: 0.3063
F-statistic: 58.17 on 2 and 257 DF,  p-value: < 2.2e-16
```

```
--- Does SP500_Returns Granger-cause M2 ? (bivariate VAR) ---
$Granger

        Granger causality H0: SP500_Returns do not Granger-cause M2

data:  VAR object subvar
F-Test = 1.5714, df1 = 2, df2 = 116, p-value = 0.2121
```

```
--- Does SP500_Returns Granger-cause M2 ? (bivariate VAR) ---
$Granger

        Granger causality H0: SP500_Returns do not Granger-cause M2

data:  VAR object subvar
F-Test = 2.0016, df1 = 1, df2 = 514, p-value = 0.1577
```

VI (H_{VI} regressions with quarterly data)

```
Estimation results for equation SP500_Returns:
==============================================
SP500_Returns = SP500_Returns.l1 + Real_GDP.l1 + M2.l1 + const

                 Estimate Std. Error t value Pr(>|t|)
SP500_Returns.l1  0.04979    0.06275   0.794   0.4282
Real_GDP.l1      -0.01966    0.47842  -0.041   0.9673
M2.l1             0.19201    0.43774   0.439   0.6613
const             1.70630    1.00306   1.701   0.0901 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Residual standard error: 8.004 on 256 degrees of freedom
Multiple R-Squared: 0.003569,   Adjusted R-squared: -0.008108
F-statistic: 0.3057 on 3 and 256 DF,  p-value: 0.8213


Estimation results for equation Real_GDP:
=========================================
Real_GDP = SP500_Returns.l1 + Real_GDP.l1 + M2.l1 + const
```

```
Estimation results for equation Real_GDP:
=========================================
Real_GDP = SP500_Returns.l1 + Real_GDP.l1 + M2.l1 + const

                 Estimate Std. Error t value Pr(>|t|)
SP500_Returns.l1 0.033354   0.007542   4.422 1.44e-05 ***
Real_GDP.l1      0.087478   0.057505   1.521    0.129
M2.l1            0.303504   0.052614   5.768 2.30e-08 ***
const            0.097309   0.120563   0.807    0.420
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Residual standard error: 0.9621 on 256 degrees of freedom
Multiple R-Squared: 0.1884,     Adjusted R-squared: 0.1789
F-statistic: 19.81 on 3 and 256 DF,  p-value: 1.409e-11
```

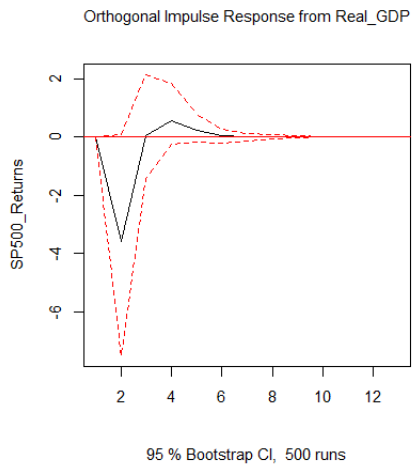
```
Estimation results for equation M2:
===================================
M2 = SP500_Returns.l1 + Real_GDP.l1 + M2.l1 + const

                  Estimate Std. Error t value Pr(>|t|)
SP500_Returns.l1 -0.010574   0.007625  -1.387    0.167
Real_GDP.l1      -0.017613   0.058139  -0.303    0.762
M2.l1             0.556495   0.053195  10.461  < 2e-16 ***
const             0.776773   0.121894   6.373 8.59e-10 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Residual standard error: 0.9727 on 256 degrees of freedom
Multiple R-Squared: 0.3119,     Adjusted R-squared: 0.3038
F-statistic: 38.67 on 3 and 256 DF,  p-value: < 2.2e-16
```

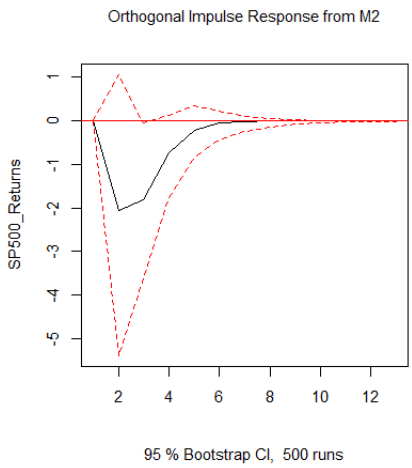
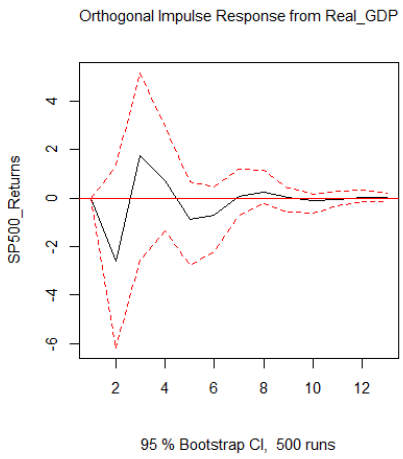
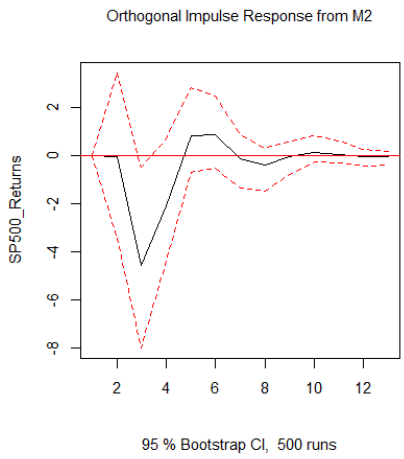
VII (H_{VI} regression, IRF, and FEVD with yearly data)

```
Estimation results for equation SP500_Returns:
==============================================
SP500_Returns = SP500_Returns.l1 + Real_GDP.l1 + M2.l1 + const

                  Estimate Std. Error t value Pr(>|t|)
SP500_Returns.l1  0.007065   0.130321   0.054 0.956945
Real_GDP.l1      -2.088890   0.975356  -2.142 0.036290 *
M2.l1            -0.636517   0.555716  -1.145 0.256592
const            19.303158   4.910935   3.931 0.000222 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Residual standard error: 15.87 on 60 degrees of freedom
Multiple R-Squared: 0.1004,     Adjusted R-squared: 0.05546
F-statistic: 2.233 on 3 and 60 DF,  p-value: 0.09357
```

FEVD of SP500_Returns

horizon	from_SP500_Returns	from_Real_GDP	from_M2
1	1.00	0.00	0.00
2	0.94	0.05	0.02
3	0.93	0.05	0.03
4	0.92	0.05	0.03
5	0.92	0.05	0.03
6	0.92	0.05	0.03
7	0.92	0.05	0.03
8	0.92	0.05	0.03
9	0.92	0.05	0.03
10	0.92	0.05	0.03
11	0.92	0.05	0.03
12	0.92	0.05	0.03





VIII (IRFs H_{VI} with yearly data and forced number of lags $k = 2$)